

Eco-Loop Building Agents

Autonomous AI for Building Energy Optimization

Physical AI & Smart Sustainable Building Operations

PROBLEM STATEMENT ID

Question 1 · PS-101 · HONEYWELL-Q1

THEME

Physical AI & Smart Sustainable Building Operations

PS CATEGORY

Software (AI & Cyber-Physical Simulation)

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The Idea

Proposed Solution — Eco-Loop Autonomous Building Agent

An autonomous closed-loop controller that pairs EnergyPlus building-physics simulation with open-source LLMs via the Model Context Protocol — running a 5-stage Simulate → Feedback → Reason → Control → Audit cycle to set thermostat and lighting schedules with zero human intervention.

16.6%

total energy reduction
verified over 168 hrs

HOW IT SOLVES THE PROBLEM



Overcomes Rigid Static Schedules

Replaces fixed timer rules with proactive, demand-response intelligence.



Carbon & Forecast Aware

Evaluates live grid carbon intensity and 6-hour weather forecasts to pre-cool spaces and harvest daylight.



Proven 16.6% Energy Reduction

259 kWh saved, 118.4 kg CO2 avoided over 168 hrs — 100% of occupant hours stayed within ISO 7730 comfort.

WHAT MAKES IT NOVEL



Automated Forward Injection

Actively rewrites runtime building models (optimized_day1 → final.idf) instead of just displaying recommendations.



Protocol Standardization (MCP)

Decouples building engineering from the AI model — plug-and-play across Gemini, Llama 3, Mistral, and Qwen.



Explainable, Zero-Downtime Audit

An offline rules engine keeps control alive through network drops; every decision is logged and explainable.



Technical Approach

TECH STACK



Core Language

Python 3.10+ unifying physics, MCP server & AI cognition.



Simulation Sandbox

EnergyPlus V26.1 + DOE Small Office model (Chicago TMY3).



Comms Protocol

MCP via FastMCP — 8 typed tools for sensors & setpoints.



Cognitive Engine

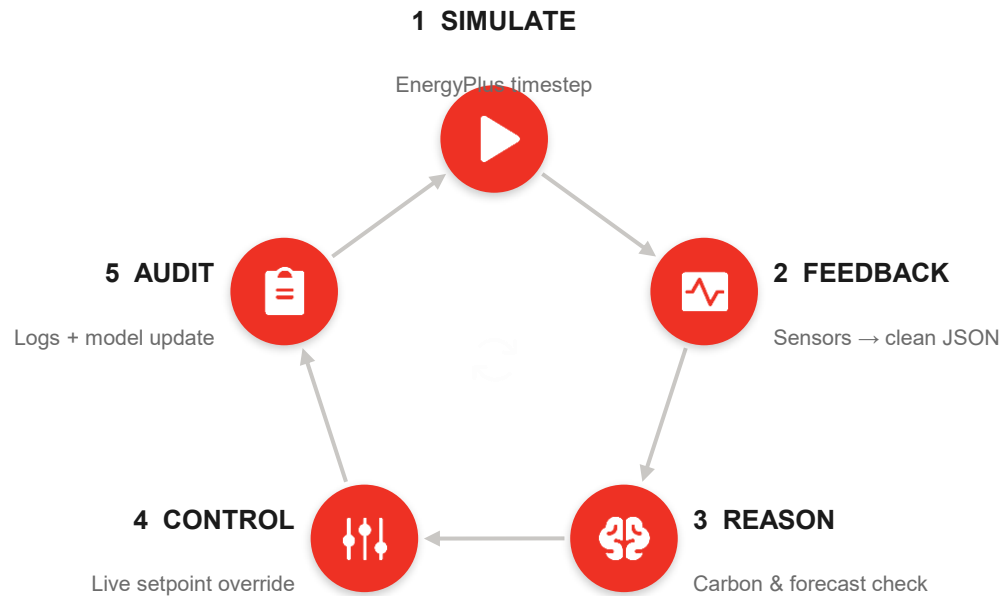
Gemini 2.0 Flash, modular with Llama 3 / Mistral / Qwen.



Analytics Dashboard

Plotly Dash + Bootstrap, live at 127.0.0.1:8050.

5-STAGE REAL-TIME EXECUTION CYCLE



WORKING PROTOTYPE VERIFICATION

1,764

autonomous MCP tool calls

168 hrs

continuous simulated runtime (7 days)

0%

crash / timeout rate



Feasibility & Viability

WHY IT'S FEASIBLE



Proven Physical AI Prototype

1,764 real-time tool calls executed across a simulated week, zero crashes.



Quantifiably Superior Economics

16.6% net energy reduction (259 kWh), 118.4 kg CO2 eliminated in one week.



Zero Hardware Upgrade Friction

Runs as a software layer over existing BMS via MCP — no controller swap.



Comfort Compatibility

100% adherence to ISO 7730 (PMV -0.5 to +0.5) — savings, no discomfort.

KEY RISKS



Cloud LLM Latency & Connectivity

API delays or outages could cause commands to miss control timesteps.



Sensor Noise & Bad Telemetry

Real thermostats/occupancy sensors may drift or miscalibrate over time.



Token Inflation & Hallucination

Long simulations can bloat context windows, risking erratic decisions.



Cybersecurity & Override Safety

Direct HVAC access risks unauthorized setpoint tampering or extremes.

OUR MITIGATIONS



Offline Fallback Engine

SmartRuleAgent takes over within 50ms of any API or network disruption.



Edge Open-Source LLM Modularity

Drop-in local models (Llama, Mistral, Qwen) via vLLM/Ollama — no cloud tie.



Sliding Window Prompt Mgmt

6-turn history cap + concise sensor summaries prevent token bloat.



Thermodynamic Safety Clamping

Hard guardrails in the MCP server bound temperatures to 18–28°C, always.

Artifacts



Live Repository

Full codebase — orchestrator, MCP server, agent logic.

github.com/SHARVESH-005/eco-loop-agent



Building Models

Base + 8 AI-modified .idf models (forward injection).



Simulation Datasets

168-hr baseline vs. optimized tracks + decision audit log.



Architecture Docs

Tool schemas, latency mitigation, LLM compatibility.



Live Optimization Dashboard

```
Problems  Output  Debug Console  Terminal  Ports

Tool: adjust_lighting_schedule({"zone_name": "Zone_East_Perimeter", "fraction": 0.1})
Tool: set_thermostat_setpoint({"zone_name": "Zone_South_Perimeter", "heating_setpoint": 18.0, "cooling_setpoint": 28.0, "fraction": 0.1})
Tool: adjust_lighting_schedule({"zone_name": "Zone_South_Perimeter", "fraction": 0.1})
Tool: set_thermostat_setpoint({"zone_name": "Zone_West_Perimeter", "heating_setpoint": 18.0, "cooling_setpoint": 28.0, "fraction": 0.1})
Tool: adjust_lighting_schedule({"zone_name": "Zone_West_Perimeter", "fraction": 0.1})
Tool: set_thermostat_setpoint({"zone_name": "Zone_Core", "heating_setpoint": 18.0, "cooling_setpoint": 28.0, "fraction": 0.1})
Tool: adjust_lighting_schedule({"zone_name": "Zone_Core", "fraction": 0.1})
Tool: log_decision({"reasoning": "Zone_North_Perimeter: Unoccupied (hour 21), setback to 18-28C;  })

Applied 10 control actions
Day 7 | Energy: 2.2 kWh | Savings: 259.0 kWh (16.6%) | LLM calls: 126

AI-Optimized simulation complete
Modified runtime .idf models saved to E:\Honeywell\eco-loop-agent\models
Total LLM calls: 126
Total tool calls: 1764
Saved to E:\Honeywell\eco-loop-agent\data\optimized_results.csv

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FINAL RESULTS: BASELINE vs AI-OPTIMIZED
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```

Metric	Baseline	AI-Optimized	Savings
Total Energy (kWh)	1558.3	1299.3	16.6%
HVAC Energy (kWh)	464.3	374.7	19.3%
Lighting Energy (kWh)	462.0	292.6	36.7%
Carbon (kg CO ₂)	711.6	593.2	16.6%

```
Total Energy Savings: 259.0 kWh (16.6%)
Carbon Reduction: 118.4 kg CO2
```

Terminal — Final Verification Run



Research & References



Live GitHub Repository

Unified codebase, building models, dashboard, and architecture docs.

github.com/SHARVESH-005/eco-loop-agent



Building Physics & Energy Simulation

DOE / NREL EnergyPlus V26.1; Small Office Reference Building, Chicago TMY3.

energyplus.net



Communication Protocol & Agent Engineering

Model Context Protocol (MCP) via FastMCP server architecture.

modelcontextprotocol.io



Human Thermal Comfort Standards

ISO 7730 & ASHRAE 55 — PMV / PPD thermal comfort indices.

iso.org/standard/39155.html



Cognitive LLM Engine & Modularity

Gemini 2.0 Flash tool-calling; open-source Llama 3 / Mistral / Qwen via vLLM/Ollama.

ai.google.dev · ollama.com

Thank you to Honeywell for organizing the OA Hackathon 2026 — we look forward to driving the next generation of autonomous, sustainable building technology.